ESSENTIALLY CONSTANT DIAGRAMS IN A KAROUBIAN CATEGORY SPLIT OFF A CONSTANT DIRECT SUMMAND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the filtered case in both directions; the closing sentence 'The proof of (2) is dual.' is the author's own duality remark on a proof written out in full inside the counted span, preserved verbatim, and no other step of the argument is declined.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Essentially constant diagrams. Categories, Definition categories-definition-essentially-constant-diagram. Supplies the object X with the maps X -> M_i and M_i -> X whose composites are eventually the identity, which the proof uses both to produce the direct summand decomposition and to verify essential constancy.
- Cofinal subcategories and essential constancy. Categories, Lemma categories-lemma-cofinal-essentially-constant. Allows replacing the index category by the cofinal (respectively initial) subcategory when checking that a diagram is essentially constant.
- Karoubian categories. Categories, the definition of a Karoubian (pseudo-abelian) category. Supplies the splitting of the idempotent needed to write M_{i'} as a direct sum of the image of X and the kernel of the map to X.

Source reference. Original definition 05PU: sources/stacks-categories/pinned-original-source.tex, source lines 3145–3145. Navigation pointer invoked at proof line 7713; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0A1S: sources/stacks-categories/pinned-original-source.tex, source lines 3439–3439. Navigation pointer invoked at proof line 7733; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{I}$ be a category, let $\mathcal{A}$ be a pre-additive Karoubian category, and let $M : \mathcal{I} \to \mathcal{A}$ be a diagram.*

(1) *Assume $\mathcal{I}$ is filtered. The following are equivalent*
   (a) *M is essentially constant,*
   (b) *$X = \operatorname{colim} M$ exists and there exists a cofinal filtered subcategory $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \mathrm{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ such that $X_{i'}$ maps isomorphically to X and $Z_{i'}$ to zero in $M_{i''}$ for some $i' \to i''$ in $\mathcal{I}'$.*

(2) *Assume $\mathcal{I}$ is cofiltered. The following are equivalent*
   (a) *M is essentially constant,*
   (b) *$X = \lim M$ exists and there exists an initial cofiltered subcategory $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \mathrm{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ such that X maps isomorphically to $X_{i'}$ and $M_{i''} \to Z_{i'}$ is zero for some $i'' \to i'$ in $\mathcal{I}'$.*

Proof. Assume (1)(a), i.e., $\mathcal{I}$ is filtered and M is essentially constant. Let $X = \operatorname{colim} M_i$. Choose i and $X \to M_i$ as in Categories, Definition 05PU. Let $\mathcal{I}'$ be the full subcategory consisting of objects which are the target of a morphism with source i. Suppose $i' \in \mathrm{Ob}(\mathcal{I}')$ and choose a morphism $i \to i'$. Then $X \to M_i \to M_{i'}$ composed with $M_{i'} \to X$ is the identity on X. As $\mathcal{A}$ is Karoubian, we find a direct summand decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$, where $Z_{i'} = \operatorname{Ker}(M_{i'} \to X)$ and $X_{i'}$ maps isomorphically to X. Pick $i \to k$ and $i' \to k$ such that $M_{i'} \to X \to M_i \to M_k$ equals $M_{i'} \to M_k$ as in Categories, Definition 05PU. Then we see that $M_{i'} \to M_k$ annihilates $Z_{i'}$. Thus (1)(b) holds.

Assume (1)(b), i.e., $\mathcal{I}$ is filtered and we have $\mathcal{I}' \subset \mathcal{I}$ and for $i' \in \mathrm{Ob}(\mathcal{I}')$ a direct sum decomposition $M_{i'} = X_{i'} \oplus Z_{i'}$ as stated in the lemma. To see that M is essentially constant we can replace $\mathcal{I}$ by $\mathcal{I}'$, see Categories, Lemma 0A1S. Pick any $i \in \mathrm{Ob}(\mathcal{I})$ and denote $X \to M_i$ the inverse of the isomorphism $X_i \to X$ followed by the inclusion map $X_i \to M_i$. If j is a second object, then choose $j \to k$ such that $Z_j \to M_k$ is zero. Since $\mathcal{I}$ is filtered we may also assume there is a morphism $i \to k$ (after possibly increasing k). Then $M_j \to X \to M_i \to M_k$ and $M_j \to M_k$ both annihilate Z_j. Thus after postcomposing by a morphism $M_k \to M_l$ which annihilates the summand Z_k, we find that $M_j \to X \to M_i \to M_l$ and $M_j \to M_l$ are equal, i.e., M is essentially constant.

The proof of (2) is dual. □